Evaluation of ECMWF S2S Forecasts over Ireland for Multi-Hazard Applications

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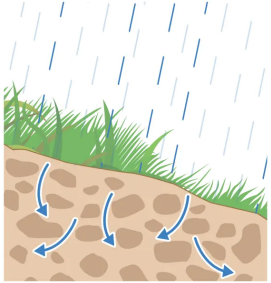
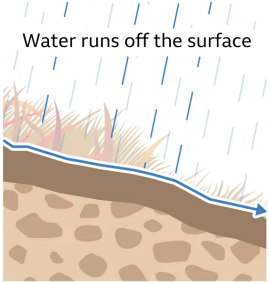
What Happens When Drought Becomes Flood?

Normal conditions



Soil absorbs water like a sponge

Drought conditions



Hard layer of soil repels water

Adapted from BBC Research (2022)

€500M+

Irish agricultural losses, 2018
Beast from the East + summer drought

Noone et al. (2025)

1.4–8×

worse impacts when
drought precedes flood

Worou & Messori (2025)

25%

of all floods globally
preceded by drought

Matanó et al. (2024)

Has Anyone Tried to Forecast This?

Research landscape: drought–flood prediction

	Retrospective	Forecast	Projection
Hazards treated independently	Drought and flood inventories (SSI, HANZE, EM-DAT, ...)	Operational flood and drought products (GloFAS, EFAS, EDO, ...)	Single-hazard climate projections (CMIP6, UKCP18, CORDEX, ...)
Hazards treated as connected	**57 papers** DFAA Whiplash D-to-F UK hydro-haz. Weather whip.	★	Steensen et al. (2025) Tan et al. (2023) Collet et al. (2018) Formetta et al. (2019) Visser-Quinn et al. (2019)

Could We Forecast It?

Trigger is identifiable

53%

of drought terminations via IVT under NAO+

Parry et al. (2023)

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Preconditions growing

$\sim 2\times$

compound preconditions increasing by mid-century

Steensen et al. (2025)

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Ireland is where they propagate

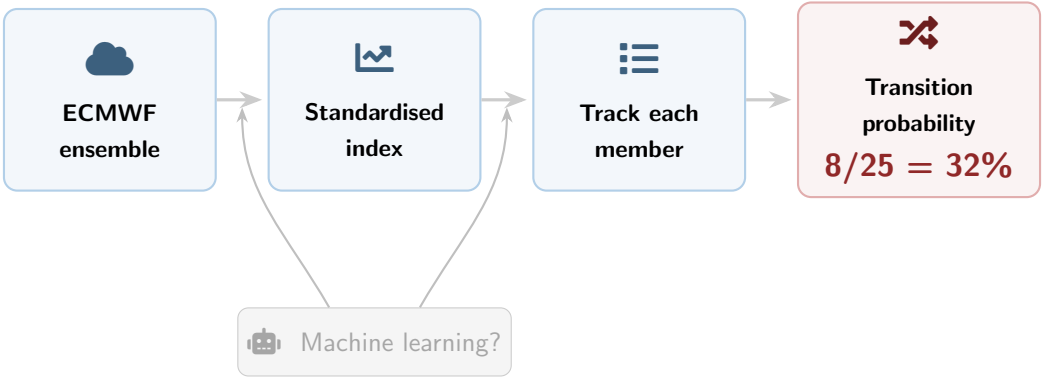
1 in 5

propagate to hydrology

$>$40% in Ireland

Brunner et al. (2025)

The Approach



What Should You Take Away?

Transition preconditions are increasing

Ireland is where they propagate

Can S2S forecasts detect them?

References

Brunner, M.I., Anderson, B., Muñoz-Castro, E. (2025). Meteorological and hydrological dry-to-wet transition events are only weakly related over European catchments. *Environ. Res. Lett.*, 20(8), 084013.

He, X., Sheffield, J. (2020). Lagged compound occurrence of droughts and pluvials globally over the past seven decades. *Geophys. Res. Lett.*, 47(14), e2020GL087924.

Matanó, A. et al. (2024). Compound and consecutive drought–flood events at a global scale. *Environ. Res. Lett.*, 19(6), 064048.

Noone, C. et al. (2025). The Beast from the East and drought of summer 2018: compound event impacts upon Ireland's agricultural sector. *Weather*, 80(2), 56–61.

Parry, S. et al. (2023). Abrupt drought termination in the British–Irish Isles driven by high atmospheric vapour transport. *Environ. Res. Lett.*, 18(10), 104050.

Steensen, B.M. et al. (2025). Future increase in European compound events where droughts end in heavy precipitation. *npj Clim. Atmos. Sci.*, 8(1), 267.

Worou, K., Messori, G. (2025). Compounding droughts and floods amplify socio-economic impacts. *Environ. Res. Lett.*, 20(10), 104024.